SYSTEM STUDY AND REPORT

Study of a Supermarket Billing and Management System

Product and Barcode Management | Billing and Invoice Generation
Discounts, GST and Totals | Cash, Card and UPI Payments
Receipt Printing | Inventory and Stock Management

Project type	System study and report
Domain	Retail information systems (supermarket)
Context	Indian retail, GST-compliant billing
Date	October 2026
Submitted by	_____

Table of Contents

1. Introduction and Background	**3**
2. System Overview and Architecture	**4**
3. Product and Barcode Management	**6**
4. Billing and Invoice Generation	**7**
5. Discounts, GST and Total Calculation	**9**
6. Cash, Card and UPI Payments	**12**
7. Receipt Printing	**13**
8. Inventory and Stock Management	**15**
9. Data Design, Reports and Security	**17**
10. Advantages of the System	**18**
11. Limitations of the System	**18**
12. Suggestions for Improvement	**19**
13. Conclusion	**20**
14. References and Further Reading	**20**

List of Tables and Figures

Tables and figures are numbered by chapter (for example, Table 5.3 is the third table in Chapter 5). Figures include a photograph of a supermarket sales floor (1.1), the architecture diagram (2.1), the end-to-end sale flow (2.2), billing steps (4.1), calculation order (5.1), dynamic UPI flow (6.1), the sample receipt (7.1) and the stock cycle (8.1).

1. Introduction and Background

1.1 Abstract

A supermarket is a high-volume retail outlet that sells thousands of different products to hundreds or even thousands of customers every day. Each sale involves identifying items, applying prices, discounts and taxes, accepting payment, issuing a receipt and reducing the stock on hand. Doing this manually with price tags, calculators and paper registers is slow, error-prone and nearly impossible to audit. A **Supermarket Billing and Management System (SBMS)** automates this complete chain of activities by combining barcode scanners, a billing terminal, a payment interface, a receipt printer and a central database.

This report is a system study of such a software system. It examines how products and barcodes are managed, how bills and invoices are generated, how discounts, Goods and Services Tax (GST) and totals are calculated, how cash, card and UPI payments are handled, how receipts are printed, and how inventory and stock are controlled. The report closes with the advantages and limitations of the system and a set of practical suggestions for improvement.

1.2 Background

Retail in India has moved quickly from traditional kirana stores to organised supermarkets and hypermarkets. A typical neighbourhood supermarket stocks between 5,000 and 15,000 distinct items (stock keeping units, or SKUs), while a large store can exceed 30,000. Prices change often, promotions run every week, and tax rules differ by product category. Customers expect a queue to move in under two minutes and expect a correct, itemised, GST-compliant bill every single time.

Before computerisation, cashiers read the price from a printed sticker, keyed it into a mechanical register or calculator, and wrote the item in a bill book. Stock was counted by hand at the end of the week or month. Typical problems were mis-keyed prices, missed discounts, wrong tax, long queues, theft that went unnoticed, and shelves that were empty before anyone noticed. Computerised billing addresses these problems at their source by making the product database the single source of truth.

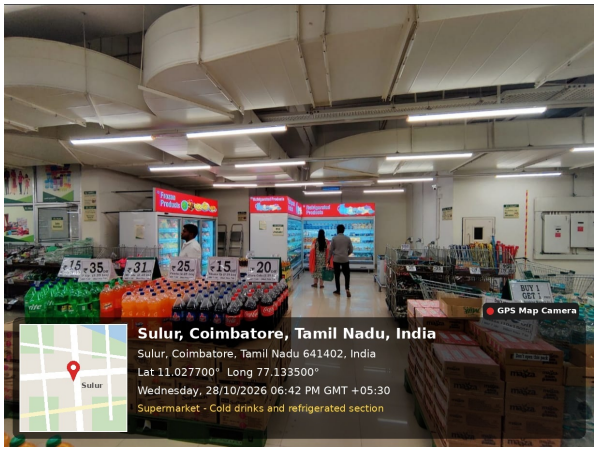


Figure 1.1: A supermarket sales floor with cold-drink displays and refrigerated sections (illustrative photograph)

1.3 Problem Statement

Supermarket owners need a system that can bill customers quickly and accurately, comply with GST invoicing rules, accept every payment mode customers use today, and keep the stock record in step with real sales. The study therefore asks: *what functions must such a system provide, how do those functions work together, and where can the design be improved?*

1.4 Objectives of the Study

- To understand the structure and working of a supermarket billing and management system.

- To study product master data and barcode handling, including how a scan becomes a bill line.
- To study invoice generation, discount rules, GST computation and bill totalling with worked examples.
- To study cash, card and UPI payment handling, including split payments and reconciliation.
- To study receipt printing formats and requirements, and inventory and stock control.
- To identify advantages, limitations and realistic suggestions for improvement.

1.5 Scope and Method of Study

The study is a **functional system study**. It describes the modules, data and workflows of a typical single-store supermarket system in the Indian context. It is based on analysis of how point-of-sale (POS) systems are commonly designed, the structure of GST tax invoices, and standard retail practice. Figures, product names, prices and invoice numbers used in the examples are illustrative and are not taken from any real store. The study does not cover supplier portals, e-commerce integration or multi-branch consolidation in depth, although these are touched upon in the improvement suggestions.

2. System Overview and Architecture

An SBMS has two faces. The **front counter** (point of sale) is used by cashiers to bill customers. The **back office** is used by the manager, store keeper and accountant to maintain products, prices, stock, suppliers and reports. Both faces read and write the same database.

2.1 Users and Roles

Role	Main responsibilities	Typical access
Cashier	Scan items, apply allowed discounts, take payment, print receipt, hold or resume bills	Billing screen only
Store keeper	Receive goods, enter purchases, print barcode labels, adjust and count stock	Products, purchase, stock
Manager	Set prices and offers, approve returns and overrides, view sales and stock reports	All modules, approvals
Accountant	GST returns, day-end cash reconciliation, supplier payments	Reports, tax, payments
Administrator	Create users, set permissions, configure printers, taxes and store details	System settings

Table 2.1: Users and roles in a supermarket billing system

2.2 Major Modules

The system is easiest to understand as seven cooperating modules. Each is studied in its own chapter of this report.

Module	Purpose	Chapter
Product and barcode management	Maintain item master, prices, tax category and barcodes	3
Billing and invoice generation	Create bills from scanned items and number invoices	4
Discount, GST and totals	Apply offers, split tax, round off and total the bill	5
Payment handling	Accept cash, card, UPI and mixed payments	6
Receipt printing	Produce the customer copy and store copy of the bill	7
Inventory and stock management	Track purchases, sales, returns, wastage and reorder levels	8
Reports and security	Sales, tax and stock reports, user rights, audit log	9

Table 2.2: Functional modules of the system

2.3 Layered Architecture

Most billing systems follow a layered design. The presentation layer is the screen the cashier or manager sees. The business layer holds the rules (pricing, discount, tax, stock deduction). The data layer holds the database. Hardware devices are connected through drivers at the edge of the system. Separating the layers means that a change in tax law, for example, touches only the business rules and not the screens or the printer code.

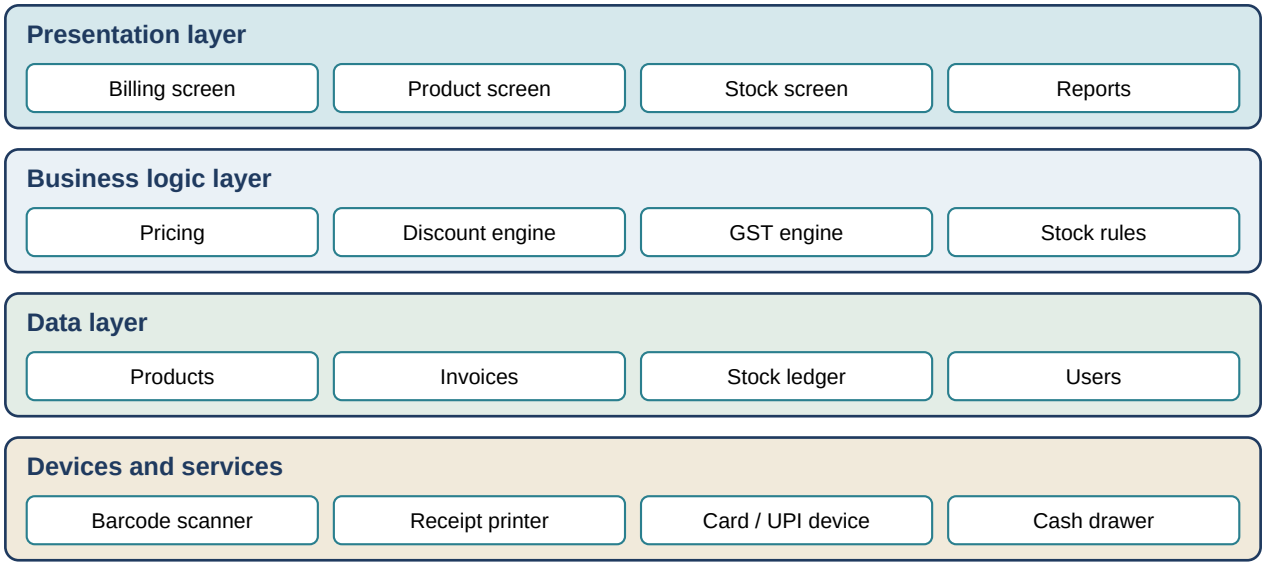


Figure 2.1: Layered architecture of a supermarket billing and management system

2.4 Typical Hardware and Software

- **Billing terminal:** a desktop or all-in-one PC with touch or keyboard, kept at each counter.
- **Barcode scanner:** a hand-held or presentation (on-counter) scanner that behaves like a keyboard.
- **Thermal receipt printer:** usually 80 mm paper width, with an auto-cutter.
- **Cash drawer:** opened automatically by the printer when a cash sale is completed.
- **Payment devices:** a card machine (POS terminal) and a UPI QR code or dynamic QR display.
- **Weighing scale:** optional, for loose items; may print its own barcode label.
- **Database server:** a local server or a cloud database, with daily backup.
- **UPS:** a power backup, because a power cut in the middle of a bill must not lose the transaction.

2.5 Overall Workflow

The life of a typical sale is shown below. Every arrow in the diagram is automated; the cashier only scans, confirms and takes payment.

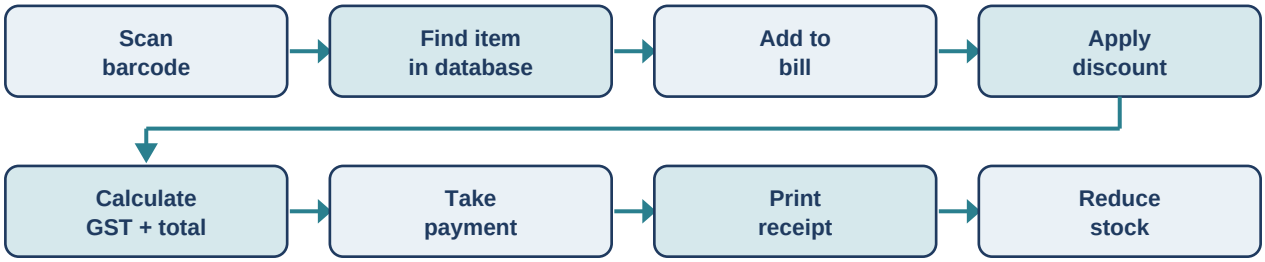


Figure 2.2: End-to-end flow of a sale

3. Product and Barcode Management

Everything in a billing system depends on the **product master**: the table that describes every item the store can sell. If the master data is wrong, every bill, tax figure and stock report built on it will also be wrong. For this reason product management is treated as the foundation module.

3.1 Product Master Data

Each product is stored as one record with a unique internal code. The fields below are the minimum a GST-compliant supermarket needs.

Field	Description	Example
Product code	Unique internal ID (primary key)	P00123
Barcode	EAN-13 or store-generated code; one product may have several	8901234567890
Name	Short name that fits on the receipt line	Toor Dal 1 kg
Category / brand	For reports, shelf layout and offers	Pulses / Local brand
Unit of measure	Piece, kg, gram, litre, pack	Pack
MRP	Maximum retail price printed on the pack (tax inclusive)	Rs. 165.00
Selling price	Price charged by the store (not above MRP)	Rs. 158.00
Purchase price	Cost price from supplier, used for margin reports	Rs. 141.00
HSN code	Harmonised classification used on GST invoices	0713
GST rate	Tax rate applicable to the item (0 / 5 / 18 / 40 %)	5 %
Reorder level	Stock quantity at which a purchase alert is raised	20
Batch / expiry	Needed for food, dairy and medicines	B245 / 03-2027
Status	Active, discontinued, blocked	Active

Table 3.1: Product master fields

3.2 Barcodes: What They Are and How They Work

A barcode is a machine-readable pattern of bars and spaces that encodes a number. The scanner shines a light or laser across the pattern, measures the width of the bars, converts them into digits and sends the digits to the computer exactly as if they had been typed. The billing software then looks that number up in the product table and returns the name and price. The barcode itself carries *no price*; it only identifies the product. This is why a price change needs only one database edit rather than relabelling every pack.

Common barcode types in supermarkets

Type	Digits	Use in a supermarket
EAN-13	13	Standard code printed on packaged goods; first digits 890 indicate India (GS1 India)
UPC-A	12	Mostly on imported goods from North America
EAN-8	8	Small packs where a 13-digit code will not fit
Code 128	Variable	Store-printed labels for loose items, batches and shelf tags
QR code	Variable	Digital offers, UPI payment display, loyalty card

Table 3.2: Barcode symbologies used in retail

An EAN-13 code consists of a country or GS1 prefix, the manufacturer code, the product code and a final **check digit**. The check digit lets the scanner detect a misread. It is calculated as follows: the digits in odd

positions (1st, 3rd, ... 11th) are added; the digits in even positions (2nd, 4th, ... 12th) are added and multiplied by three; the two results are summed; and the check digit is the number needed to bring the total up to the next multiple of ten.

> **Worked example: check digit for 890123456789?**
> Odd positions: 8+0+2+4+6+8 = 28. Even positions: 9+1+3+5+7+9 = 34, and 34 x 3 = 102. Total = 28 + 102 = 130. The next multiple of ten is 130 itself, so the check digit is **0**, giving 8901234567890.

3.3 Adding and Maintaining Products

New products enter the system in three ways. First, the store keeper can **add a product manually** by scanning the pack's existing barcode into the barcode field and typing the remaining details. Second, a large supplier list can be **imported from a spreadsheet** (CSV or Excel) with columns matching Table 3.1. Third, items with no barcode, such as loose vegetables, bakery items or repacked grains, are given a **store-generated code** and a printed label. The system checks that a barcode is not already assigned to another product, because a duplicate would make the scanner return the wrong item.

Price and tax edits are controlled. Only a user with manager rights may change a selling price, and the system stores the old value, the new value, the user and the time in an audit table. Products that are discontinued are marked inactive rather than deleted, so that old invoices still show the correct name.

3.4 Multiple Barcodes, Weighed Items and Variants

- **Multiple barcodes per product:** the same item may arrive with different manufacturer batches or pack designs. A linking table allows several barcodes to point to one product code.
- **Weighed items:** a scale prints a label whose barcode embeds either the item code and the weight or the item code and the price. The system splits the number, recognises the item and takes the quantity from the label, so the cashier does not need to weigh again.
- **Pack variants:** a 500 g and a 1 kg pack of the same brand are separate products with separate barcodes, stock and prices, but share a brand and category.
- **Combo packs:** a bundle is stored as a parent product whose components are deducted from stock when the bundle is sold.

3.5 Barcode Label Printing

For items that arrive without a barcode, or for shelf-edge price tags, the system prints labels on a label printer. The user selects products or a goods-receipt note, chooses a label size, and the software prints one label per unit, showing the name, price, barcode and optionally the date of packing. Using the same database for labels and billing guarantees that the price on the shelf, the price on the screen and the price on the receipt are identical.

4. Billing and Invoice Generation

Billing is the heart of the system and the part customers see. The objective is to convert a basket of physical goods into a correct, numbered, legal document in a few seconds per item, and to do so without error even at peak hour.

4.1 The Billing Screen

The billing screen is designed around speed. The cursor stays in the barcode field so that each scan adds a line without any keystroke. The main area lists the scanned items with quantity, rate, discount and amount. A large panel on the side shows the running total, which is the figure the customer watches. Function keys or touch buttons are provided for the common actions.

Action	Purpose	Control
Scan / enter item	Add one unit of the item; repeated scan increases quantity	Open to all cashiers
Change quantity	Set quantity for bulk purchases or weighed goods	Open to all cashiers
Delete line	Remove a mistaken scan	Logged; manager PIN above a limit
Hold bill	Park the bill so the next customer can be served	Open to all cashiers
Recall bill	Resume a held bill	Open to all cashiers
Item / bill discount	Reduce price by percent or amount	Within the cashier's limit; else manager PIN
Customer	Attach mobile number or loyalty card	Optional
Cancel bill	Void the whole transaction	Manager PIN; always logged

Table 4.1: Common billing actions and their controls

4.2 Steps in Generating a Bill

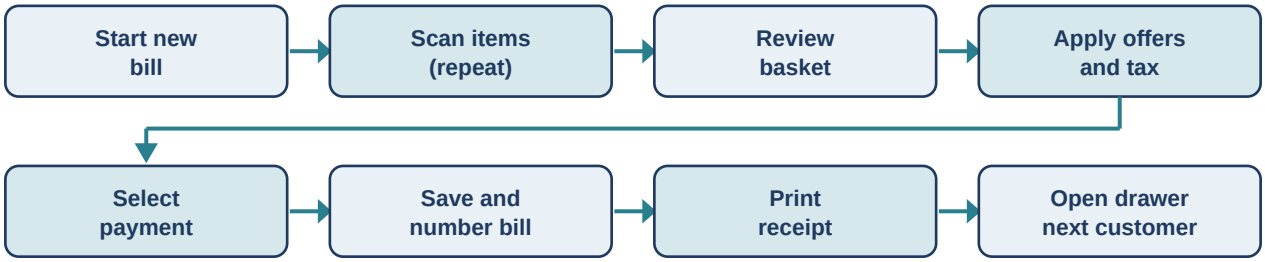


Figure 4.1: Steps in bill generation

While the cashier is scanning, the bill exists only in memory as a *draft*. It receives its legal invoice number only at the moment payment succeeds. This avoids wasting numbers on abandoned bills and keeps the invoice series continuous, which tax authorities expect.

4.3 Invoice Numbering and Mandatory Content

Under GST every supply must be supported by an invoice. For retail sales to consumers, the invoice must have a unique serial number for the financial year (maximum 16 characters, using letters, digits, hyphen or slash). The system therefore generates numbers automatically in a format such as **SMK/2627/004582**, where SMK is the store code, 2627 is the financial year 2026-27 and the last six digits are the running sequence. The sequence restarts on 1 April. Two terminals must never issue the same number, so the sequence is allocated by the database rather than by each terminal.

A GST retail invoice normally contains the following information:

- Supplier name, address and GSTIN; invoice number and date and time of sale.
- Description of each item, HSN code (as required for the store's turnover), quantity and unit.
- Rate, taxable value, discount, and tax rate with the amounts of CGST and SGST (or IGST).
- Total invoice value, in figures, after any round-off.
- Place of supply and state code, and, where required, customer name and GSTIN for business buyers.

> **Note on compliance**
> Tax rules, rate slabs and the invoice details required for a given turnover are revised by notification. The system should keep these as configurable settings and the store should verify them with its tax adviser or the official GST portal before go-live.

4.4 Retail Invoice and Business (B2B) Invoice

Most supermarket sales are to ordinary consumers and use a short retail invoice. When a customer such as a restaurant, school or small business wants to claim input tax credit, the cashier enters the customer's GSTIN and the system prints a full tax invoice carrying the buyer's name, GSTIN and address. The GSTIN is validated by format and checksum at entry. The same sale data is used; only the document layout and the tax report in which it is counted differ.

4.5 Holding, Returns and Cancellations

A **held bill** lets a cashier serve another customer when the first has forgotten something. A **sales return** is never done by editing or deleting the original invoice. Instead the system creates a **credit note** that refers to the original bill number, lists the returned items, reverses the tax and puts the goods back into stock (or into a damaged-goods location). A **cancelled** bill keeps its number but is marked void with the reason and the approving user, so the number series has no gaps.

4.6 Safeguards During Billing

- Prices come only from the product master; the cashier cannot type a rate unless given the right.
- Selling price may not exceed the MRP recorded for the item.
- Every line deletion, discount, price override and cancellation is recorded with user and time.
- A bill is written to the database in one atomic transaction, so a power failure leaves either a complete bill or none.
- Terminals can continue in an offline mode with a local copy of products and sync later.

5. Discounts, GST and Total Calculation

This module turns the list of scanned items into the final amount the customer pays. The order of calculation matters: discounts reduce the value on which tax is computed, and tax is computed before the bill is rounded. If the order is wrong the customer is over- or under-charged and the GST return will not match.

5.1 Types of Discount

Type	How it works	Example
Item discount (percent)	A fixed percent off a product for a period	10 % off biscuits
Item discount (amount)	A fixed rupee amount off	Rs. 5 off on a 1 kg pack
Buy X get Y	Free or reduced units on quantity bought	Buy 2 get 1 free
Combo / bundle price	A set of products at a special total price	Tea + biscuits at Rs. 99
Bill-value discount	Discount when the basket crosses a limit	Rs. 50 off above Rs. 1,000
Loyalty / membership	Points or a standing percent for members	2 % member discount
Payment offer	Discount for a specific card or UPI app	5 % off with a partner card
Coupon / voucher	A code entered or scanned at billing	WELCOME50

Table 5.1: Discount types supported by the discount engine

A **discount engine** stores each offer with its start and end date, the products or categories it applies to, a priority, and whether it may be combined with others. At billing time the engine evaluates all active offers on the basket and applies the best allowed combination. Discounts that are given *before* the supply and are shown on the invoice reduce the taxable value. A discount allowed by a later credit note is treated differently, which is one more reason to let the software, not the cashier, decide.

5.2 GST Basics for Billing

GST is a destination-based tax on supply of goods and services. When a supermarket sells within its own state the tax is split equally into **CGST** (central) and **SGST** (state, or UTGST in a union territory). For a sale to a buyer in another state it is charged as **IGST**. A store billing walk-in customers over the counter almost always uses CGST plus SGST.

India's rate structure was simplified with effect from 22 September 2025, following the 56th GST Council meeting. The earlier 12 % and 28 % slabs were removed, leaving 5 % and 18 % as the main rates, a special 40 % rate for sin and luxury goods such as aerated and sweetened drinks and tobacco, and nil rates for essentials. Because tax rates change, they are stored in a **tax table** linked to each product and carrying an effective-from date, so that historic invoices keep their original rate and new sales use the new one.

GST rate	Supermarket examples (illustrative)	CGST + SGST
0 % (nil)	Fresh vegetables and fruit, milk, unbranded staples as notified	0 + 0
5 %	Many packaged foods, edible oil, soaps, toothpaste, shampoo, hair oil	2.5 % + 2.5 %
18 %	Most other goods: electrical items, many appliances, some processed items	9 % + 9 %
40 %	Aerated and sweetened beverages, tobacco products, pan masala	20 % + 20 %

Table 5.2: GST rate slabs and examples (verify against current notifications)

5.3 Tax-Inclusive and Tax-Exclusive Pricing

In a supermarket the printed **MRP is inclusive of all taxes**, so the customer pays the shelf price and the tax is already inside it. The system therefore has to work backwards from the tax-inclusive price to find the taxable value:

Formulae

Taxable value = Net amount / (1 + GST rate). GST amount = Net amount - Taxable value. CGST = SGST = GST amount / 2.

For a B2B price quoted *excluding* tax: GST = Taxable value x Rate, and Invoice value = Taxable value + GST.

5.4 Order of Calculation

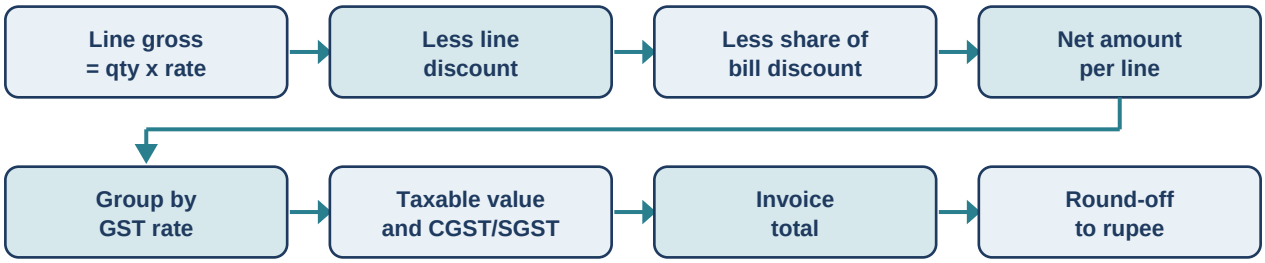


Figure 5.1: Order of calculation from line value to payable amount

5.5 Worked Example

The bill below is a small, fictitious basket. Prices are tax-inclusive. The biscuits carry a 10 % line discount. Taxes are then worked out slab by slab, which is how the GST summary at the foot of a retail invoice is normally presented.

#	Item	Qty	Rate	Gross	Disc.	Net	GST %
1	Toor Dal 1 kg	2	158.00	316.00	0.00	316.00	5
2	Biscuits 100 g	3	30.00	90.00	9.00	81.00	5
3	Shampoo 180 ml	1	209.60	209.60	0.00	209.60	5
4	Cold Drink 600 ml	2	40.00	80.00	0.00	80.00	40
5	LED Bulb 9 W	1	120.00	120.00	0.00	120.00	18
6	Tomato (loose)	1.350 kg	40.00	54.00	0.00	54.00	0
	Total			**869.60**	**9.00**	**860.60**	

Table 5.3: Line items of the sample bill (Rs.)

GST %	Net (incl. tax)	Taxable value	CGST	SGST	Total tax
0	54.00	54.00	0.00	0.00	0.00
5	606.60	577.71	14.45	14.44	28.89
18	120.00	101.69	9.16	9.15	18.31
40	80.00	57.14	11.43	11.43	22.86
Total	**860.60**	**790.54**			**70.06**

Table 5.4: GST summary of the sample bill (Rs.)

The net amount is Rs. 860.60. Rounded to the nearest rupee the invoice carries a round-off of +Rs. 0.40, so the **amount payable is Rs. 861.00**. Of the net amount, Rs. 70.06 is GST and Rs. 790.54 is the taxable value. Where the tax on a slab is an odd number of paise, the extra paisa is given to SGST so that CGST plus SGST always equals the total tax.

5.6 Rounding and Precision

Money is stored as an exact decimal or as an integer number of paise, never as a binary floating-point number, because floating point cannot represent values such as 0.10 exactly and errors accumulate over thousands of lines. Tax is calculated per slab and rounded once rather than per item, which keeps the invoice total identical to the sum of its parts. The round-off is shown as a separate line and is posted to its own ledger account.

5.7 Common Calculation Errors That the System Prevents

- Applying the discount after tax instead of before.
- Charging tax on top of an MRP that already includes it.
- Using the old tax rate after a rate change.
- Rounding every line separately so that the bill total differs from the tax summary by a few paise.
- Selling above MRP by manual price entry.
- Stacking two offers that were never meant to be combined.

6. Cash, Card and UPI Payments

A bill is complete only when it is paid. The payment module records how the customer paid, ties the payment to the invoice, and feeds the end-of-day cash and bank reconciliation. A modern supermarket must accept at least three modes: cash, debit or credit card, and UPI.

Aspect	Cash	Card	UPI
Confirmation	Cashier counts notes	Terminal approval code	App success message and callback
Typical speed	Fast, but change-making takes time	10-20 seconds	5-15 seconds
Cost to store	Handling and bank-deposit risk	Merchant discount rate on card value	Generally nil or very low for small merchants (check current rules)
Settlement	Same day in the drawer	Next working day to bank	Usually same or next day to bank
Main risk	Counterfeit notes, theft, counting errors	Chargebacks, skimming	Wrong-account confusion, delayed confirmation
Data stored by billing system	Amount tendered, change	Last 4 digits, approval code, RRN	UPI transaction ID, payer app

Table 6.1: Comparison of payment modes

6.1 Cash Payments

The cashier selects *Cash* and keys the amount received, or taps a note denomination. The system computes the **change due** (amount tendered minus amount payable) and displays it in large figures. On saving, it commands the printer to open the cash drawer. For the above sample bill of Rs. 861.00, a customer handing over Rs. 1,000 would receive Rs. 139.00 in change. The system refuses to accept an amount tendered that is less than the due amount unless another mode covers the balance.

At the end of the shift the cashier counts the drawer. The system prints a **cash-up report** that shows opening float, cash sales, cash refunds and paid-outs, and the expected closing cash. The difference from the counted cash is recorded as short or excess against the cashier.

6.2 Card Payments

Card payments are processed on a separate payment terminal supplied by a bank or payment company. In a **standalone** setup the cashier types the amount on the card machine, and then records the approval code in the billing screen. In an **integrated** setup the billing software sends the amount to the terminal over USB, Ethernet or the network, and the terminal returns the result automatically, removing keying errors.

The billing system must **never store the full card number, expiry date, PIN or CVV**. It keeps only the masked number (last four digits), the card network, the approval code and the retrieval reference number. This keeps the store clear of the heaviest data-security obligations under card industry standards.

6.3 UPI Payments

Unified Payments Interface (UPI) lets a customer pay from a bank account using any UPI app by scanning a QR code or entering a UPI ID. Two QR styles are used in stores:

- **Static QR:** a fixed code on the counter. The customer types the amount. The cashier must check the success message on the shop's own phone or soundbox, as a customer's screenshot is not proof.
- **Dynamic QR:** the billing system generates a code carrying the exact bill amount and a unique reference. The customer cannot mistype, and the system receives a confirmation from the bank or payment gateway and marks the bill as paid automatically. This is the preferred approach.

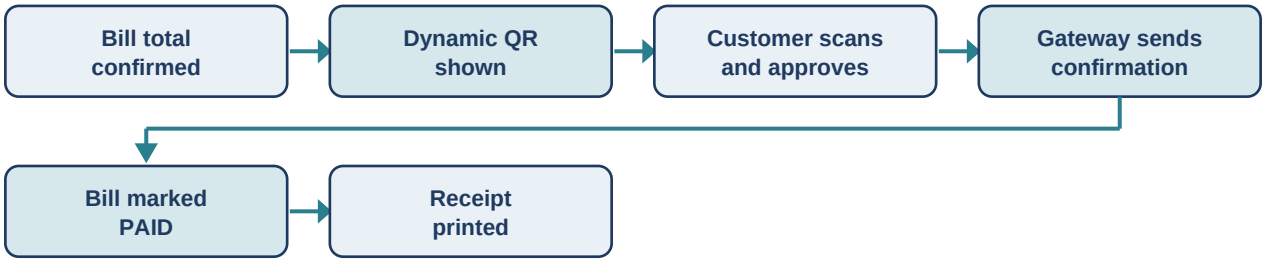


Figure 6.1: Dynamic UPI payment flow

If the confirmation does not arrive in about a minute, the system shows the status as *pending* and offers a *check status* button instead of asking the customer to pay again. This avoids double payment, which is the commonest UPI problem at a counter.

6.4 Split and Mixed Payments

Customers often pay part in cash and part by UPI or card. The payment screen therefore keeps a list of payment lines until the total equals the amount payable. For example, the sample invoice of Rs. 861.00 could be settled as Rs. 500.00 cash and Rs. 361.00 by UPI. Each line is stored separately with its own mode and reference, and the daily report adds them by mode.

6.5 Failed Payments, Refunds and Reconciliation

- **Declined card or failed UPI:** the bill stays open; the cashier chooses another mode.
- **Refund:** given in the original mode where possible. A card refund is reversed through the terminal, a UPI refund goes back to the payer's account, and a cash refund needs manager approval.
- **Reconciliation:** at day end, card and UPI totals in the billing system are compared with the bank or gateway settlement statements. Differences are listed for investigation.
- **Audit trail:** every payment line holds the cashier, terminal, time and external reference.

7. Receipt Printing

The receipt is the customer's proof of purchase, the legal copy of the invoice, and the document needed for any return or warranty claim. It is produced within a second or two of payment, and its layout must be clear enough to be checked at the counter.

7.1 Printer Technology

Supermarkets use **direct thermal printers**. These have no ink or ribbon; a heated print head darkens heat-sensitive paper. They are quiet, fast (200 mm per second or more), and need little maintenance. The two common paper widths are 58 mm for small shops and **80 mm** for supermarkets, which fits about 42 to 48 characters per line. A built-in auto-cutter tears off the bill, and a drawer-kick port opens the cash drawer. Thermal printouts fade with heat and light over time, so stores keep the digital copy in the database and can reprint it on demand.

Printer feature	Recommendation	Reason
Paper width	80 mm	Room for item name, quantity, rate and amount on one line
Speed	200 mm/s or faster	Keeps queues moving at peak hours
Cutter	Auto-cutter	Saves seconds on every bill
Interface	USB plus Ethernet or Bluetooth	Flexibility in counter layout
Cash drawer port	RJ11 drawer kick	Opens the drawer on cash sales

Printer feature	Recommendation	Reason
Command set	ESC/POS compatible	Widely supported by billing software

Table 7.1: Receipt printer specification

7.2 Receipt Layout

A receipt is divided into a header, a body of item lines, a totals block, a tax summary and a footer. The sample bill worked out in Chapter 5 is printed below in the form a thermal printer would produce. The store name, address and GSTIN are sample values.

```
          SMK FRESH SUPERMARKET
  12, Market Road, Coimbatore - 641 001
     GSTIN: 33ABCDE1234F1Z5 (sample)
               TAX INVOICE
----------------------------------------
Inv No : SMK/2627/004582
Date   : 28-10-2026   Time: 18:42
Cashier: C02          Counter: 3
----------------------------------------
Item                Qty    Rate     Amount
----------------------------------------
Toor Dal 1 kg         2  158.00     316.00
Biscuits 100 g        3   30.00      90.00
  Discount                           -9.00
Shampoo 180 ml        1  209.60     209.60
Cold Drink 600 ml     2   40.00      80.00
LED Bulb 9 W          1  120.00     120.00
Tomato (loose)     1.35   40.00      54.00
----------------------------------------
Items: 6                  Gross     869.60
                       Discount      -9.00
                      Round off      +0.40
NET PAYABLE                Rs.     861.00
----------------------------------------
GST   Taxable    CGST    SGST
0%       54.00     0.00     0.00
5%      577.71    14.45    14.44
18%     101.69     9.16     9.15
40%      57.14    11.43    11.43
----------------------------------------
Paid: CASH 500.00  UPI 361.00
You saved Rs. 9.00 today
         Thank you. Visit again!
     Returns within 7 days with bill
```

Figure 7.1: Sample 80 mm thermal receipt generated from the worked example

7.3 Contents of the Receipt

Section	Content
Header	Store name, logo, address, phone, GSTIN, the words TAX INVOICE
Bill details	Invoice number, date, time, counter, cashier, customer name or mobile if captured
Item lines	Item name, quantity, rate, discount line (if any), line amount
Totals	Item count, gross value, total discount, round-off, net payable
Tax summary	Rate-wise taxable value, CGST and SGST
Payment	Mode(s), amount tendered, change returned, reference numbers
Footer	Savings message, return policy, loyalty points, barcode or QR of the invoice number

Table 7.2: Sections of a receipt

7.4 Printing Process

The billing software builds the receipt as formatted text or as a small drawing, then sends it to the printer either through the operating system's print driver or directly with ESC/POS commands. Direct commands

are faster and let the program select bold text, double-height totals, centred headings, barcodes and a paper-cut command. A reprint of an old bill is marked **DUPLICATE** so that it cannot be mistaken for the original.

7.5 Alternatives to Paper

To reduce paper use, the same invoice can be sent to the customer as a PDF or link by SMS, WhatsApp or email, using the mobile number entered at billing. Customers who need no paper receive only a digital copy; the store still holds the full record in its database. This also lets customers view their purchase history and lowers printing cost.

8. Inventory and Stock Management

Inventory is the money the store has tied up in goods. Too little stock means lost sales and unhappy customers; too much stock means cash locked up, expiry losses and storage cost. The inventory module keeps the stock record accurate, so that the store buys the right quantity at the right time.

8.1 Stock Movements

Stock changes only through documented movements. The system keeps a **stock ledger** in which every movement adds one row (product, date, type, quantity, reference). Current stock is the sum of the ledger, so any figure can be traced to the documents that created it.

Movement	Effect on stock	Source document
Purchase receipt	Increase	Goods receipt note (GRN) and supplier invoice
Sale	Decrease	Sales invoice (automatic at billing)
Sales return	Increase (or damaged stock)	Credit note
Purchase return	Decrease	Debit note to supplier
Wastage / damage / expiry	Decrease	Stock adjustment with reason
Stock count difference	Increase or decrease	Physical verification entry
Transfer between locations	Decrease at source, increase at destination	Transfer note

Table 8.1: Stock movement types

8.2 Purchase and Goods Receipt

The cycle starts when stock falls below the reorder level. The store keeper raises a **purchase order** to a supplier. When the goods arrive they are checked against the delivery challan and the order, and a **GRN** is entered. For each item the person scans the barcode, enters the quantity, batch, expiry date, purchase price and MRP. The system updates stock, updates the cost price and calculates the input GST. A difference between ordered and received quantities is highlighted so that it can be raised with the supplier.

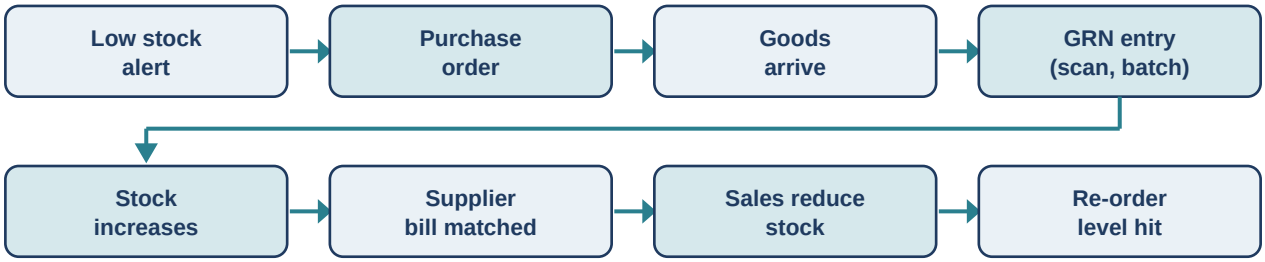


Figure 8.1: Purchase-to-sale stock cycle

8.3 Real-Time Stock Update at Billing

When a bill is saved, each line writes a negative row to the stock ledger in the same database transaction. Stock is therefore always current. If a cashier tries to sell a quantity that the system thinks is not in stock, the program can warn, ask for a manager PIN or block the sale, depending on store policy. A warning is usually preferred because the shelf may hold stock that was never received in the system, and it flags the gap for correction.

8.4 Reorder Levels and Stock Valuation

The **reorder level** is the stock at which a new order must be placed. A simple formula is: reorder level = average daily sales x supplier lead time in days + safety stock.

> **Worked example**
> A pack of tea sells 12 units a day on average. The supplier delivers in 3 days and the store wants 2 days of safety stock (24 units). Reorder level = 12 x 3 + 24 = **60 units**. When the stock drops to 60, the system raises a low-stock alert.

Stock is valued for the balance sheet by a consistent method. **FIFO** (first in, first out) assumes the oldest purchases are sold first and is natural for food. **Weighted average cost** recalculates the unit cost after each purchase. Whichever is chosen, it is applied consistently so that gross profit reports are meaningful.

8.5 Batch and Expiry Control

Dairy, bakery, packaged food and medicines must be sold before they expire. The system stores stock **by batch** with an expiry date, sells the earliest-expiring batch first (FEFO), shows a list of items expiring within a chosen number of days, and blocks the sale of expired items at the counter. Near-expiry stock can be moved onto an automatic markdown offer through the discount engine.

8.6 Stock Taking and Adjustments

Even the best system drifts from reality through theft, miscounting, damage and unrecorded returns. Stock is therefore counted physically at intervals. A **cycle count** checks a few categories each week instead of closing the store for a full count. Staff scan and count each item; the system lists the variance between book and physical stock; a manager reviews and approves adjustments with a reason code. The variance trend is useful, because a repeated shortage in one category often points to theft or a pricing or barcode error.

8.7 Inventory Reports

Report	Question it answers
Current stock	How much of each item is on hand now, and what is its value?
Low-stock / reorder	Which items must be ordered today?
Expiry report	Which batches expire in the next 7, 15 or 30 days?
Fast and slow movers	Which items sell most, and which have not sold for 60 or 90 days?
Stock ledger	How did the stock of one item arrive at its present level?
Shrinkage report	How much was lost to damage, wastage or count differences?
Margin report	What profit does each product or category earn?

Table 8.2: Key inventory reports

9. Data Design, Reports and Security

9.1 Principal Database Tables

A relational database holds the information of the system. The main tables and their key links are given below. Each invoice line refers to a product, and each stock-ledger row refers to the document (invoice, GRN, adjustment) that caused it.

Table	Key fields	Linked to
Product	product_id, name, category, unit, MRP, price, HSN, gst_rate, reorder_level	Barcode, Stock
Barcode	barcode, product_id	Product
Invoice	invoice_no, date_time, cashier_id, customer_id, gross, discount, tax, round_off, total, status	Invoice_Line, Payment
Invoice_Line	invoice_no, product_id, qty, rate, discount, taxable, cgst, sgst	Invoice, Product
Payment	invoice_no, mode, amount, reference, time	Invoice
Stock_Ledger	product_id, batch, date, type, qty, ref_doc	Product
Supplier / Purchase	supplier_id, GRN no, date, product lines, cost, GST	Product, Stock_Ledger
User / Audit	user_id, role, action, old_value, new_value, time	All tables

Table 9.1: Principal database tables

9.2 Management Reports

- **Daily sales:** bills, items, average bill value, sales by hour and by cashier.
- **Payment-mode summary:** cash, card and UPI totals for reconciliation.
- **GST report:** rate-wise taxable value and tax collected, supporting preparation of returns.
- **Discount report:** offers used, amount given and who approved overrides.
- **Category and brand sales:** to plan shelf space and negotiate with suppliers.
- **Returns and cancellations:** volume and reasons, an early warning for fraud or training needs.

9.3 Security and Data Protection

- **Role-based access:** each user sees only the functions that the role needs.
- **Strong login:** individual user IDs, password or PIN; shared logins are avoided.
- **Audit log:** price changes, deletions, voids and overrides cannot be removed by ordinary users.
- **Backups:** automatic daily backup to a separate device or the cloud, with a restore test.
- **Encryption and network safety:** database access limited to the store network; remote access through a secured connection only.
- **Customer data:** mobile numbers and purchase history are personal data and are collected with consent and protected as required by India's data protection law.

10. Advantages of the System

Area	Advantage	Benefit to the store
Speed	Scanning replaces typing; totals are instant	Shorter queues, more customers served per hour
Accuracy	Prices, discounts and tax come from the database	Fewer billing errors and disputes
GST compliance	Numbered tax invoices and rate-wise summaries	Simpler returns and fewer notices
Payments	Cash, card, UPI and split payments in one screen	Convenience for every customer
Inventory	Real-time stock and reorder alerts	Fewer stock-outs, less dead stock
Control	Roles, PINs, audit trail	Lower risk of fraud and pilferage
Information	Sales, margin and stock reports	Better buying and pricing decisions
Customer service	Itemised receipt, loyalty offers, easy returns	Trust and repeat visits
Cost	Less paper, less manual accounting	Lower running cost over time

Table 10.1: Advantages and their business benefit

10.1 Detailed Discussion

Accuracy and trust. A customer is likely to notice if an item is billed above the shelf price. Because the shelf label, the screen and the receipt are all generated from one price list, the three agree. When a promotion ends the discount stops automatically on the end date, with no need to remember to remove it at every counter.

Better decisions. The sales and margin reports let the owner see which products bring profit rather than only turnover. Items that sell slowly can be discounted or dropped, and fast movers can be stocked deeper. Peak-hour data helps decide how many counters to staff.

Accountability. Every sale is tied to a cashier and a terminal. Cash differences, unusual voids and heavy discounting by one user become visible in the exception reports. This tends to reduce losses simply because staff know the actions are recorded.

11. Limitations of the System

No system is perfect. The limitations below are not reasons to avoid computerisation, but they must be understood so that they can be planned for and reduced.

Limitation	Explanation	Possible impact
Dependence on power and hardware	Scanner, PC, printer or network failure stops the counter	Queues, lost sales
Initial cost	Hardware, software, installation, label printing and training	Hurdle for small stores
Data entry effort	Thousands of products must be entered accurately at start	Delays at launch
Master-data errors	A wrong price, GST rate or barcode spreads into every bill	Over/under-charging, tax errors
Staff training	Cashiers and keepers need training; resistance to change	Errors in early weeks

Limitation	Explanation	Possible impact
Barcode problems	Damaged, missing or duplicated barcodes	Slow scanning, wrong item billed
Stock mismatch	Theft, unrecorded returns and breakage make book stock differ from shelf	Wrong reorder decisions
Payment dependency	UPI and card need bank and network availability	Failed or pending payments
Cybersecurity	Unpatched PCs and weak passwords can expose data	Data loss, fraud, ransomware
Rule changes	GST rates and invoice rules change from time to time	Needs timely software update
Limited analytics	Basic systems give reports but little forecasting	Missed optimisation

Table 11.1: Limitations and their impact

11.1 Discussion

The most serious practical limitation is **dependence on correct master data**. The software multiplies whatever is stored: a single wrong GST rate on a fast-selling item can distort tax for a whole month before it is noticed. Regular master-data audits and approval for rate and price edits are essential.

A second limitation is **operational fragility**. A paper bill book works in a power cut; a computer does not. Stores need a UPS, a spare scanner, a backup printer and a written offline procedure for the occasional long outage. Finally, the system captures what is *recorded*, not what actually happens on the shelf, so physical stock counts remain necessary however good the software.

12. Suggestions for Improvement

The suggestions below are arranged from quick, low-cost changes to larger projects, so that a store can improve in stages.

12.1 Short-Term Improvements (0 to 3 months)

- **Dynamic UPI QR integration** with automatic payment confirmation, replacing the static code.
- **Integrated card terminal** to remove manual entry of approval codes.
- **Barcode validation** with a check-digit test and duplicate alert at the time of entry.
- **Tax table with effective dates** so that rate changes need no program change.
- **Automatic daily backup** to cloud storage with weekly restore test.
- **Cashier training and a short quick-reference card** at every counter.

12.2 Medium-Term Improvements (3 to 12 months)

- **Digital receipts** by SMS, WhatsApp or email to reduce paper cost.
- **Loyalty and CRM module** with points, customer history and targeted offers.
- **Automated purchase suggestions** based on sales speed, lead time and reorder level.
- **Expiry-based dynamic markdown** to cut wastage of dairy and bakery items.
- **Handheld devices** for stock taking and shelf label printing.
- **E-invoicing and return-filing integration** where the store's turnover requires it.
- **Offline-first terminals** that continue billing and sync later.

12.3 Long-Term Improvements (beyond 12 months)

- **Multi-store head-office consolidation** with central price and offer control.
- **Demand forecasting with machine learning** using seasonality, festivals and weather.
- **Self-checkout counters** and mobile scan-and-go billing in the store app.

- **Online ordering and home delivery** sharing the same stock and pricing database.
- **Supplier integration** for electronic purchase orders and invoices.
- **Shelf cameras or RFID** to detect empty shelves and improve stock accuracy.
- **Dashboards** for owners on mobile, with alerts for unusual voids, shortages and cash differences.

12.4 Suggested Implementation Roadmap

Phase	Time	Key activities	Expected result
1. Stabilise	0-3 months	Clean master data, dynamic UPI, backup, training	Fewer errors, safer data
2. Optimise	3-12 months	Loyalty, digital receipts, auto reorder, expiry offers	Higher sales, less wastage
3. Expand	12+ months	Multi-store, forecasting, self-checkout, online orders	Growth and competitive edge

Table 12.1: Phased improvement plan

13. Conclusion

This study examined a supermarket billing and management system from the counter to the stockroom. It showed that the system is not a single program but a set of closely linked functions that all draw on one product master. A barcode scan identifies a product; the product record supplies its price and tax rate; the discount and GST engines turn the basket into an exact invoice; the payment module records how the invoice was settled; the printer produces the customer's copy; and the stock ledger quietly reduces the inventory behind the scenes.

The advantages, namely speed, accuracy, GST compliance, flexible payments, real-time stock and managerial control, clearly outweigh the limitations. The limitations of cost, hardware and power dependence, data-entry effort, and reliance on correct master data can be managed through a UPS and backup, careful training, approval rules and periodic audits. The suggested improvements (dynamic UPI QR, integrated card terminals, digital receipts, automated reordering, expiry-based markdowns, loyalty tools and in the longer term forecasting and self-checkout) offer a practical route for a store to move from basic automation to a data-driven retail operation.

In summary, a well-designed supermarket billing and management system is the operational backbone of a modern store. Its value lies not only in faster queues but in the reliable, auditable data it creates, which lets the owner buy, price and staff the store with confidence.

14. References and Further Reading

Rules, rates and specifications change, so consult the latest official versions.

- GST Council of India, 56th Meeting press release and rate notifications, September 2025 (rate rationalisation to 5 %, 18 % and 40 % with effect from 22 September 2025). gstcouncil.gov.in
- Central Goods and Services Tax Act, 2017 and CGST Rules, 2017, particularly the rules on tax invoices, credit and debit notes (Chapter VI). Central Board of Indirect Taxes and Customs (CBIC), cbic.gov.in
- GS1 India. Barcode standards and EAN-13 guidelines. gs1india.org
- National Payments Corporation of India (NPCI). UPI product and technical documentation. npci.org.in
- PCI Security Standards Council. Payment Card Industry Data Security Standard (PCI DSS). pcisecuritystandards.org
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